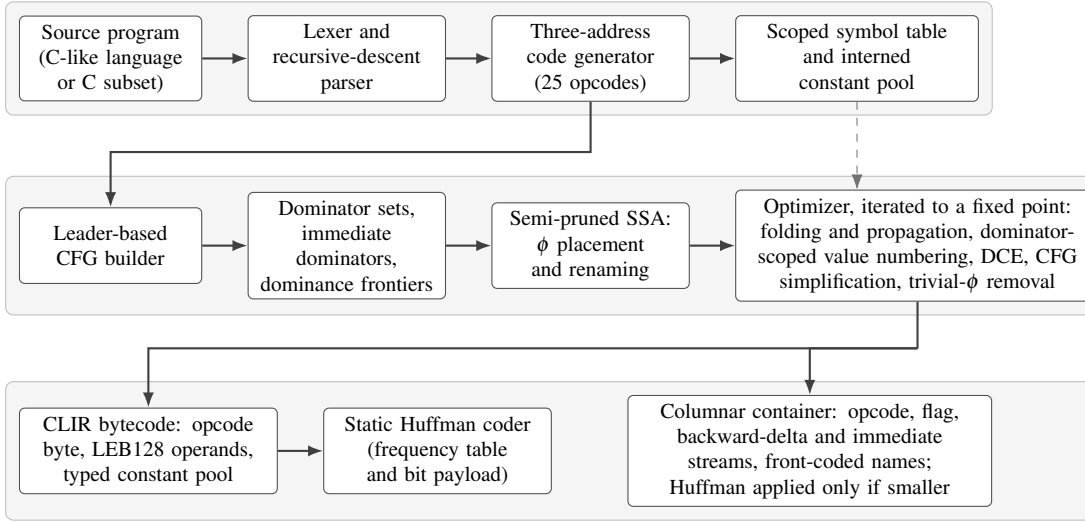
Front end
Source program (C-like language or C subset)
Lexer and recursive-descent parser
Three-address code generator (25 opcodes)
Scoped symbol table and interned constant pool
Middle end
Leader-based CFG builder
Dominator sets, immediate dominators, dominance frontiers
Semi-pruned SSA: ϕ placement and renaming
Optimizer, iterated to a fixed point: folding and propagation, dominator-scoped value numbering, DCE, CFG simplification, trivial-ϕ removal
Encodings
CLIR bytecode: opcode byte, LEB128 operands, typed constant pool
Static Huffman coder (frequency table and bit payload)
Columnar container: opcode, flag, backward-delta and immediate streams, front-coded names; Huffman applied only if smaller